Transfer Learning for Hydrological Early Warning

Predicting streamflow when local data are scarce

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1 Introduction

For most rivers, there are no long records of how much water flows through them. When only one or two years of data are available for a gauge, machine learning will memorize the data for that river instead of learning about the relationship between the river's flow and different weather patterns. Transfer learning can be used to solve this problem:

1. Train a model on **donor basins** with long records outside the watershed of the river to be measured.
2. Apply that model to the **target basin**, even if it has a short record of its flow.
3. **Fine-tune** the model with a few years of data from the target river.

The model forecasts daily streamflow and the likelihood of flow crossing drought or flood thresholds over the coming 1, 3, or 7 days. The code is open source at github.com/krishharma/hydro-tl-ews, with setup in data/README.md, settings in configs/, and run instructions in docs/RUNNING.md.

Although a full continental CAMELS-US run was put off because of compute limits, a Midwest laptop study was finished using the supervisor's advice to optimize for what this PC can handle. The target was Lick Creek near Perry, MO (USGS 05507600), with 17 similar Midwest donor basins. Results and a check against the project objectives are in Section 5. A later sensitivity pass tried a smaller donor pool, mean daily temperature, and multi-seed averaging (Section 5.7); it did not beat the 17-donor benchmark. The original Merced / multi-regime plan is still in the repository configs; basins for that larger study are listed in Table 3.

2 Data

2.1 CAMELS-US

All real experiments use **CAMELS-US**, a dataset that includes ∼671 gauges located throughout the contiguous United States. At each gauge, there are daily measurements of precipitation, temperature (from Daymet), streamflow (from USGS), and various attributes of the watershed itself. The majority of the gauges are used as **donors** (training data), while **targets** are held out for testing (Figure 1). The CAMELS dataset is not bundled with this repository, but instructions for downloading it are in data/README.md.

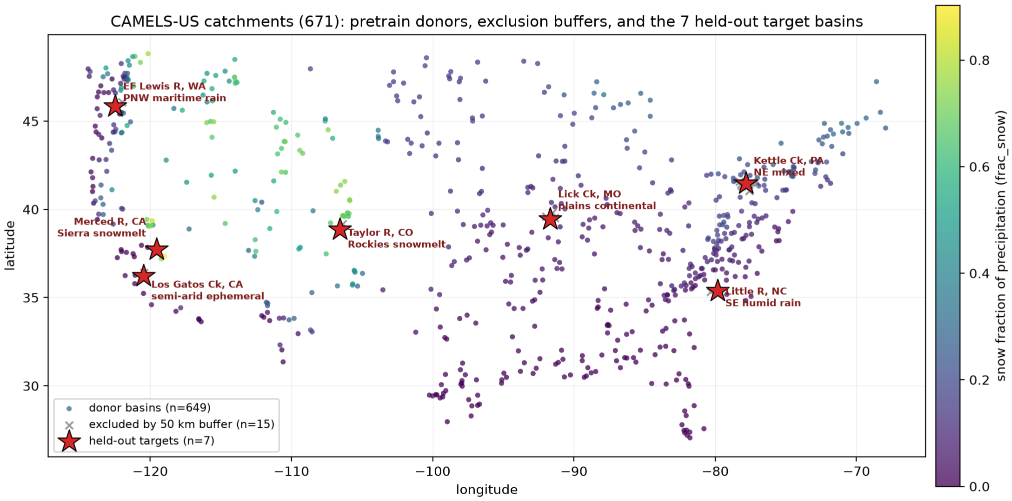

Figure 1: CAMELS-US gauge locations. Target basins (Table 3) are excluded from donor training.

2.2 Model inputs and output

Daily weather (six variables): precipitation, maximum and minimum temperature, shortwave radiation, vapor pressure, and day length.

Watershed properties (27 static attributes): topography, climate, land cover, soils, and geology. These attributes describe the watershed that is being simulated by the model. The same weather patterns can lead to different streamflow in different watersheds due to these different watershed properties.

Prediction target: daily streamflow in mm/day, derived from USGS streamflow measurements in cfs by dividing by watershed area. Table 1 summarizes all inputs.

Table 1: Summary of model inputs.

Type	Variables
Daily weather	Precipitation, $T_{\max}$, $T_{\min}$, shortwave radiation, vapor pressure, day length
Watershed	27 static attributes: elevation, slope, area, climate indices, land cover, soils, geology
Target	Daily streamflow (mm/day)

2.3 Time periods and extremes

The available local data are limited to the warmup period alone (about two years of target discharge), and the model is then tested outside that period on 2011–2014 (Table 2; walk_forward.yaml). Flood and drought levels are based on percentiles calculated over the long record from 1990–2010: **Q5** (drought), **Q95** (flood), and **Q99** (severe flood).

Table 2: Time periods used in the completed Midwest run (gauge 05507600).

Period	Dates	Purpose
Donor pretrain	Oct. 1995–Sep. 2008	Train regional model on Midwest donors
Pretrain validation	Oct. 2008–Sep. 2010	Early stopping / model selection
Target warmup	Jan. 2009–Dec. 2010	Simulate scarce local data (~2 yr)
Target evaluation	2011–2014	Fixed-window and walk-forward tests
Threshold history	1990–2010	Define Q5, Q95, Q99

Table 3: Seven target basins and climate types.

Gauge	River	Climate
11264500	Merced R., CA	Sierra snowmelt (primary)
14222500	EF Lewis R., WA	Pacific Northwest rain
01544500	Kettle Ck., PA	Mixed snow/rain
09107000	Taylor R., CO	Rockies snowmelt
02128000	Little R., NC	Humid Southeast
05507600	Lick Ck., MO	Continental plains
11224500	Los Gatos Ck., CA	Semi-arid

3 Methodology

The workflow consists of four main stages: training the model on a collection of regional basins, adapting the model to the particular basin of interest, evaluating the model on that basin using walk-forward analysis, and early-warning scoring.

3.1 Regional model (EA-LSTM)

The model that is used in this project is an **Entity-Aware LSTM** (EA-LSTM). For the EA-LSTM, the properties of the watershed control the **input gate**, while the weather and memory drive the remaining gates of the LSTM. Thus, each EA-LSTM can be trained to represent multiple basins. Each day, the EA-LSTM is provided with the weather of the previous lookback window (365 days in the full configs; **90 days** in the Midwest laptop run), and it can use this information to forecast the streamflow that will occur that day. During pre-training, the EA-LSTM is exposed to streamflow data from multiple different basins and minimizes its error in predicting discharge from each of those basins.

3.2 Transfer to the target basin

After pre-training, four adaptation strategies are compared:

- **Zero-shot** — no local training; tests the model's knowledge of the regional corpus alone.
- **Approach A** — trains only the output head of the EA-LSTM while leaving the LSTM portion intact (safest with limited local data).
- **Approach B** — trains the output head and the upper LSTM layers at a reduced learning rate (more adaptation, higher overfitting risk).
- **Local baseline** — trains a new EA-LSTM from scratch on the warmup data only.

3.3 Walk-forward evaluation

Random splits of the streamflow time series into train and test data can lead to data leakage of temporal structure because the flow on one day is similar to the flow on the previous day. Shuffled splits can therefore make models appear to perform better than they do in reality. This project uses **walk-forward evaluation** to approximate deployment with limited startup data:

1. Start with warmup data only.
2. Forecast the next block of days and score against observations.
3. Every 90 days, retrain on all discharge from the warmup start forward.
4. Apply online bias correction.
5. Advance through 2011–2014 without using future observations before forecasting them.

This evaluation method is more rigorous than a single random split of the data into train and test sets and provides a more honest estimate of performance when only a short local record exists at startup.

3.4 Early warnings and metrics

A post-processing layer translates streamflow forecasts to Q5, Q95, and Q99 crossing probabilities at 1-, 3-, and 7-day leads. Warning skill scores are based on AUC, F1@0.5, and the Brier score. These warnings are based on hindcasted flow forecasts, not a separate multi-day weather forecast. Streamflow forecast skill is scored using NSE, KGE, and PBIAS.

4 Software and status

Each stage is controlled by a YAML file in configs/. Table 4 lists the input to each stage and the output of each stage after running the full experiments.

Midwest laptop study (what was actually run): run_midwest_mini.py with configs under configs/midwest_mini/. The supervisor-optimization follow-up is run_midwest_opt.py with configs/midwest_opt/. Larger optional workflows are still in run_full_pipeline.sh and run_multi_target.py. More caveats are in KNOWN_LIMITATIONS.md.

Table 4: Software inputs and expected outputs.

Item	Description
Python 3.10+	Dependencies in requirements.txt
CAMELS-US	Placed under `data/` for real experiments
YAML config	Paths, date ranges, model settings per stage
Checkpoint	Pre-trained weights for fine-tune and evaluation
`results/midwest_mini/`	Checkpoints, metrics, predictions for completed run
`walk_forward.csv`	Daily observed and predicted flow
`walk_forward_metrics.json`	NSE, KGE, PBIAS, warning scores
`walk_forward_warnings.csv`	Event labels and probabilities
`summary.json`	Aggregated Midwest study metrics

5 Results and observations

5.1 What was actually run

The original project brief asked for pretraining on all ∼671 CAMELS-US basins and then transferring to a snowmelt-dominated Sierra/Rockies target. After the supervisor suggested focusing on what could be finished on this PC, the completed experiment used Midwest data only:

- **Hardware:** Apple M4 laptop, Apple Silicon MPS, ∼16 GB RAM.
- **Data extract:** CAMELS attributes for all gauges, but only Daymet forcings and USGS streamflow for HUC folders `04`, `05`, `07`, and `10` (∼200 MB of time series, not the full ∼14 GB continental unpack).
- **Geographic pool:** 163 CAMELS basins inside the Midwest box [36.5°N, 49.5°N]×[−104°W, −80.5°W].
- **Donors:** the 17 basins most similar to the target in static attributes, after a 50 km exclusion buffer and dropping gauges that were missing local files.
- **Target:** USGS **05507600** (Lick Creek near Perry, MO; continental plains; frac_snow ≈ 0.07).
- **Model:** EA-LSTM, hidden size **128**, lookback **90** days, dropout 0.4.
- **Pretrain:** 10 epochs, batch size 128, NSE loss; validation NSE-loss fell from 0.94 (epoch 1) to 0.78 (epoch 10).
- **Adaptation compared:** zero-shot; Approach A (head-only fine-tune on the 2-year warmup); local-from-scratch baseline. Approach B was not run here.
- **Walk-forward:** 2011–2014, refit every 90 days (**17 refits**), `refit_train_start=2009-01-01` so refits stay honest to scarce local data, online bias correction, 1 461 scored days.
- **SHAP:** GradientExplainer after walk-forward (200 background / 100 sample sequences); importance written to `results/midwest_mini/shap_global_importance.csv`.
- **Command:** `python scripts/run_midwest_mini.py` (configs in `configs/midwest_mini/`).

- **Artifacts:** `results/midwest_mini/summary.json` plus the metric, prediction, warning, and SHAP files beside it.

5.2 Continuous streamflow skill

Table 5 shows NSE, KGE, and PBIAS on the shared 2011–2014 window ($n = 1461$ days). Transfer from similar Midwest stations beats training from scratch on the two-year warmup alone. Fine-tuning also raises KGE and nearly clears the large negative volume bias seen in zero-shot. Walk-forward is the strongest continuous result here, which fits a model that is refit as local observations arrive.

Table 5: Midwest mini-study continuous skill (USGS 05507600, 2011–2014).

Setting	NSE	KGE	PBIAS (%)
Zero-shot (pretrained only)	0.150	0.032	−37.5
Fine-tune, head only (Approach A)	0.160	0.112	+4.1
Local baseline (from-scratch warmup)	0.009	−0.351	+72.6
Walk-forward (17× 90-day refits)	0.179	0.161	−8.7

Absolute NSE values (∼0.15–0.18) sit below published large-sample LSTM numbers from models trained on hundreds of donors with longer sequences. That gap is expected for a 17-donor, 90-day laptop setup on a flashier plains hydrograph. What still matters for this project is the ranking: **transfer** $\gg$ **local**.

5.3 Early-warning skill

Thresholds from the long local record were Q5 = 0.0 mm/d (near-zero / intermittent low flow at this gauge), Q95 = 3.26 mm/d, and Q99 = 16.91 mm/d. Table 6 summarizes walk-forward warning scores. Flood and drought discrimination (AUC) is strong to good, especially at short leads. F1 at probability 0.5 stays modest, and Brier scores get worse at 3- and 7-day leads. That pattern fits warnings built from a hindcasted daily flow model rather than a separate multi-day weather forecast. Q99 F1 is undefined because too few events cross the 0.5 probability cutoff—a rarity issue, not a broken pipeline.

Table 6: Walk-forward early-warning scores (USGS 05507600).

Event	Lead	AUC	F1@0.5	Brier
Flood Q95	1 d	0.922	0.286	0.094
Flood Q95	3 d	0.914	0.133	0.374
Flood Q95	7 d	0.908	0.219	0.693
Flood Q99	1 d	0.950	—	0.005
Flood Q99	3 d	0.931	—	0.014
Flood Q99	7 d	0.916	—	0.033
Drought Q5	1 d	0.805	0.410	0.214
Drought Q5	3 d	0.811	0.264	0.589
Drought Q5	7 d	0.819	0.276	0.803

5.4 SHAP interpretation

SHAP was also run after walk-forward on the fine-tuned Midwest model. GradientExplainer used 200 background sequences and 100 samples from the target basin history. The highest mean absolute SHAP values were all weather variables. Precipitation ranked first, then vapor pressure, shortwave radiation, day length, minimum temperature, and maximum temperature (Table 7). Static watershed attributes had mean absolute SHAP of zero in this single-basin setup. That does not mean the model ignores catchment properties; those inputs are fixed for one gauge, so within-basin attributions naturally point at the weather that changes day to day. For this rainfall-runoff plains basin, precip leading the ranking is hydrologically sensible.

Table 7: Top SHAP drivers (mean |SHAP|) for USGS 05507600.

Feature	**Mean |SHAP|**
Precipitation (mm/day)	0.0097
Vapor pressure (Pa)	0.0055
Shortwave radiation (W/m^2)	0.0052
Day length (s)	0.0041
$T_{\min}$ (°C)	0.0036
$T_{\max}$ (°C)	0.0035
Static catchment attributes (all)	0.0000

5.5 Objective-by-objective assessment

Against the original research objectives, under the Midwest scope the supervisor approved:

1. **Regional pre-training pipeline — met (optimized).** An EA-LSTM was pretrained on real CAMELS Midwest donors. The full 671-basin continental pretrain was skipped on purpose.

2. **Fine-tuning for a data-scarce target — met.** Approach A (LSTM frozen; head trained on ∼2 years of local Q) finished on 05507600. The brief first preferred a snowmelt target; the supervisor chose Midwest plains while keeping the same scarce-data protocol.

3. **Quantify the value of transfer learning — met.** Fine-tuned transfer beat local-from-scratch by a wide margin (NSE 0.160 vs. 0.009; KGE 0.112 vs. −0.351) and improved bias versus zero-shot (PBIAS +4% vs. −37%).

4. **Walk-forward validation — met.** Rolling-origin testing with honest `refit_train_start`, 17 refits, and online bias correction gave the best continuous scores in this study (NSE 0.179, KGE 0.161).

5. **Probabilistic early warning — partially met.** AUC, F1, and Brier were produced at leads 1/3/7 for Q5, Q95, and Q99. Discrimination is strong; decision metrics (F1, longer-lead Brier) are weaker and should be read carefully.

6. **Explainable AI (SHAP) — met.** GradientExplainer attributions were computed on the fine-tuned model. Weather drivers dominate; precip is the top feature, which fits a plains rainfall-runoff basin.

5.6 Verdict

Yes — under the optimized Midwest scope, all six research objectives are addressed. The run shows regional EA-LSTM pretraining on a laptop-sized real CAMELS extract; with only about two years of local data, transfer learning beats training from scratch; walk-forward gives an honest ops-style skill estimate; flood/drought warning scores can be made from the same forecasts; and SHAP points at weather drivers led by precipitation.

This paper does *not* claim continental CAMELS performance, snowmelt-regime transfer, or a seven-basin multi-regime study. Those stay as future work if more compute and disk become available. A longer narrative is in docs/OBSERVATIONS_MIDWEST.md.

5.7 Sensitivity to further optimizations

After the Midwest pilot, the supervisor suggested a few additional changes that had shown up in other recent hydrology work. These were tested on the same Lick Creek target (USGS 05507600).

First, instead of 17 similar Midwest donors, only about eight of the nearest neighbors were kept (Pool et al., 2021, *Water Resour. Res.*). Asking for the seven nearest left only five basins after gauges missing from the partial CAMELS extract were dropped, so later runs asked for the ten nearest and kept eight with local files. Second, the LSTM forget-gate bias was already set to +3.0 in the baseline, so that change did not need to be added. Third, daily minimum and maximum temperature were replaced with a single daily mean temperature. Fourth, the same setup was trained with three random seeds (42, 123, and 456), and the daily streamflow predictions were averaged before scoring.

None of these changes beat the original 17-donor Midwest pilot on continuous skill (Table 8). For 2011–2014, walk-forward NSE was 0.179 for the baseline, 0.138 when donors were pruned but Tmin/Tmax were kept, and 0.165 for the combined mean-temperature plus three-seed ensemble with eight donors. Fine-tune NSE followed the same pattern: 0.160, 0.113, and 0.133. In every setting, transfer learning still beat training from scratch under the two-year scarce-data protocol.

Table 8: Sensitivity to supervisor optimizations (USGS 05507600, 2011–2014 NSE).

Setting	Donors	FT	Local	WF
Baseline (Tmin+Tmax, seed 42)	17	0.160	0.009	0.179
Prune only (Tmin+Tmax, seed 42)	8	0.113	0.009	0.138
Prune + mean T + 3-seed avg.	8	0.133	−0.097	0.165

So the 17-donor Midwest run remains the main Lick Creek benchmark. The follow-up is still useful as a sensitivity check: a smaller donor pool and simpler temperature inputs did not improve continuous skill here, even though they cut compute a little and are easy to defend. Full numbers are in docs/OBSERVATIONS_OPT.md; the runner is scripts/run_midwest_opt.py.